

Chih-Wei Joshua Liu

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EDUCATION

Leland Stanford Junior University Biophysics PhD Program · Thesis advisor: Grant M. Rotskoff	Stanford, CA, USA 09/2024 – present
Massachusetts Institute of Technology (MIT) Bachelor of Science in Physics – Focus Area in Chemistry GPA: 4.9/5.0 · MCAT: 528/528	Cambridge, MA, USA 05/2022
National Experimental High School SAT: 1600/1600 · ACT: 36/36	Hsinchu, Taiwan, ROC 05/2018

SELECTED PROJECTS

C.-W.J. Liu, J. Klinger, G.M. Rotskoff, Optimal parameterization of nonequilibrium generalized master equations from discrete-time experimental data, [arXiv:2606.28289](#), in revision at *Phys. Rev. X* (2026). [\[code\]](#)

- Derived the first maximum-likelihood parameterization of non-Markovian master equations validated on real experimental data and satisfying all physical constraints in general stochastic systems.
- Set new state-of-the-art among coarse-grained models in reproducing experimentally measured folding and relaxation rates of benchmark HP35-Lys24Nle/Asn27His/Lys29Nle protein from atomistic molecular-dynamics simulation.

J. Li*, **C.-W.J. Liu***, M. Szurek, and N. Fakhri, **Measuring irreversibility from learned representations of biological patterns**, *PRX Life* **2**, 033013 (2024), *equal contribution. [\[code\]](#)

- Extended thermodynamic-irreversibility estimation to entire experimental videos of high-dimensional biological processes using variational-autoencoder representations.

H. Lee*, E. Koskela*, S. Gokhale*, J. Li*, C. Fei, **C.-W.J. Liu**, L. Lin, J. Dunkel, D.J. Skinner, and N. Fakhri, **Topological flowscape reveals state transitions in nonreciprocal living matter**, *Phys. Rev. X* **16**, 031061 (2026), *equal contribution.

- Trained a classifier to distinguish particle classes with nonreciprocal interactions, enabling analyses of previously experimentally unobserved nonequilibrium states.
- Featured in *Physics* magazine and on the cover of *Phys. Rev. X*.

Change-point detection with guaranteed convergence · [py-step-detect](#)

Developed `scikit-learn`-style Python interface for detecting change points in noisy time series, implementing the current state-of-the-art dynamic-programming-based algorithm.

POSITIONS

MIT Physics of Living Systems Group	Research Assistant (06/2022 – 08/2024)
MIT Department of Physics: Teaching Assistant, 8.241 Introduction to Biological Physics (09/2023 – 05/2024); Tutor, 8.044 Statistical Physics I (02/2021 – 05/2021); Teaching Assistant, 8.01 Physics I (09/2019 – 12/2019).	
Massachusetts General Hospital	Data Analysis Intern (06/2021 – 08/2021)
Performed statistical analyses of patient data, resulting in authorship of two clinical publications.	

LEADERSHIP AND HONORS

National Science Foundation Graduate Research Fellowship (2024, Physics).
National Defense Science and Engineering Graduate Fellowship (2024, Physics; declined for GRFP).
Sigma Pi Sigma: Physics honor-society inductee and MIT Chapter President (2022); **MIT News:** [profile \(2022\)](#).
USA Biology Olympiad: Top-50 High Honorist (2017); Semifinalist (2015–2018).
MIT Society of Physics Students: President (2021–2022).
Phi Kappa Theta – Massachusetts Eta: Service Chair (2019–2020); Pledgemaster (2021). Sourced and fundraised donation of 12,000 surgical masks to Beth Israel Deaconess Medical Center during pandemic shortage.

SKILLS AND INTERESTS

Computing: lightning, jax, pydantic, numpy, scipy, torch, scikit-learn, statsmodels, uv, ty, ruff, modal
Natural language: Standard Chinese (native)
Research interests: nonequilibrium statistical mechanics, time series, optimal transport, LLM expressivity
Personal interests: phonology, historical linguistics, international-relations theory, Midwest emo, ichthyology